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AI-Driven Additive Manufacturing for Defense Readiness and Supply Chain Resilience

I. Executive Summary

No single investment in Artificial Intelligence (AI) will have a greater impact on America’s technological supremacy, national security posture, and sustained economic prosperity than investment in AI to support American manufacturing.

Based on the evolving character of warfare and heightened industrial competition, we urge the Office of Science and Technology Policy (OSTP) to include provisions in the AI Action Plan that drive insertion of AI-enabled advanced manufacturing technology within America’s public and private sector factories

Advancements in AI have enabled development of AI-integrated Additive Manufacturing (AI-AM) systems to enhance and support defense readiness, supply chain resilience, and national security. This white paper provides a recommended roadmap for implementing a strategic initiative to incorporate AI into American Additive Manufacturing and support on-demand, decentralized production of critical components. If executed, this initiative will ensure that the U.S. defense industrial base can leap ahead of global competitors in next-generation manufacturing innovation, produce hardware at the scale required to sustain high-intensity conflict scenarios, and win economic competition with strategic rivals like China.

By leveraging AI for design optimization, predictive maintenance, and production scalability, the U.S. can fortify its manufacturing sector, reduce reliance on global supply chains, and sustain a competitive advantage in emerging technological battlefields.

II. Background: The Strategic Imperative for AI-Enabled Additive Manufacturing

The National Strategy for Advanced Manufacturing, released in October 2022 by the Committee on Technology of the National Science and Technology Council, highlighted AI as a transformative force in modern industry. However, gaps in domestic manufacturing capability—particularly in defense-related sectors—remain and continue to pose a critical vulnerability. This plan, and many similar strategic documents published by DoD, congressional interests, think tanks, and academia, have highlighted several key challenges to increasing US production. These include:

- **Declining U.S. Manufacturing Capacity:** The U.S. has underinvested in industrial production for decades, leading to vulnerabilities in defense supply chains.
- **Supply Chain Disruptions:** Dependencies on overseas suppliers, particularly for high-tech components and raw materials, create national security risks.
- **Great Power Conflict and Emerging Warfare Trends:**

- The Russia-Ukraine War demonstrated that low-cost mass production of artillery shells, drones, and electronic warfare systems is essential for sustained operations.
- The Israel-Hamas Conflict underscored the effectiveness of low-cost, mass-produced drones and loitering munitions in asymmetric warfare.

These developments suggest that future conflicts will demand rapid, large-scale production of essential defense assets – a requirement that America’s legacy industrial infrastructure is ill-equipped to meet. Greater insertion of AI coupled with AI-enabled advanced manufacturing techniques can directly address these challenges by turbocharging American industry to outmatch foreign production and deter strategic rivals.

III. AI-Enabled Additive Manufacturing: A Strategic Response

The Artificial Intelligence Action Plan should include provisioning for a federal AI-integrated Additive Manufacturing (AI-AM) initiative that aligns with existing DoD, NASA, DARPA, and defense contractor efforts to integrate AI-driven smart manufacturing into the defense industrial base. This initiative should promote to the greatest extent possible integration of the following AI-enabled technologies within public and private manufacturing operations across the US. This should include both design/manufacturing of new products and an examination/potential redesign of existing and legacy products.

A. AI-Augmented Design Optimization for Performance and Production

- AI-driven generative design software tools to optimize product structures to minimize material usage and manufacturing costs while maximizing manufacturability and product performance (e.g., product strength, mass).
- Machine learning algorithms for real-time process control and quality assurance in AM facilities.
- AI-enabled software tooling to produce digital twin models and iteratively test these models against physical requirements, resulting in 3D printable computer-aided designs (CAD) that can be rapidly produced using AM printers.

B. Predictive Maintenance & Supply Chain Resilience

- AI-powered predictive analytics to identify potential component failures before they occur.
- Automated production scheduling to adapt manufacturing output to emerging defense needs.

C. AI-Integrated Manufacturing for Critical Defense Needs

- Rapid transitioning of CAD drawings to AM printers to flexibly produce defense critical structures at scale, including:
 - High-performance autonomous UAVs, hypersonic components, and space systems.
 - Sustainment components for legacy systems (to solve parts obsolescence issues).

IV. Implementation Roadmap

The Artificial Intelligence Action Plan can support the Implementation and proliferation of AI-AM through mandating a three-phase approach.

Phase 1 (2025-2027): AI Integration into Defense Manufacturing

- Call for establishment of AI-AM pilot programs within DoD and partner agencies.
- Deploy AI-driven generative design software and predictive maintenance capabilities within existing AM Facilities and Government-operated manufacturing Centers of Excellence.
- Invest in AI-powered digital twins for manufacturing process and performance optimization.

- Design and implement a compartmentalized digital infrastructure to integrate secure and reliable AI capabilities into manufacturing workflows and existing industrial control systems, defending AI-AM capabilities from foreign adversaries.

Phase 2 (2027-2029): Scaling AI-AM for Defense & Civilian Applications

- Expand decentralized AI-AM factory hubs across US to reduce reliance on foreign supply chains.
- Implement AI-driven logistics and predictive procurement to enhance DoD supply chain resilience.

Phase 3 (2029-2032): Full AI-AM Integration into U.S. Industrial Base

- Establish AI-AM as the default manufacturing process for critical defense components.
- Continue expanding decentralized AI-AM factory hubs, including to allied nations, to create a resilient allied manufacturing network that promotes rapid AI-enabled manufacturing and innovation across allied industrial bases.
- Strengthen cybersecurity measures to protect nationwide-network of AI-AM capabilities from foreign adversaries.

V. Expected Impact & Strategic Advantages

Through recommending implementation of the AI-AM initiative, the AI Action Plan will deliver benefits to the US industrial base and broader economy including:

- **Supply Chain Security:** Reduced dependency on overseas suppliers for critical military components.
- **Cost Savings:** AI-driven process efficiencies lower material and labor costs.
- **Operational Readiness:** On-demand production reduces equipment downtime and enhances force agility.
- **Economic Growth:** Strengthened domestic manufacturing base creates high-tech jobs and innovation hubs.

VI. Conclusion

AI-enabled Additive Manufacturing is not just an option—it is a strategic mandate to ensure U.S. national security and industrial resilience. Through including the AI-AM initiative in the AI Action Plan, the US Government can leverage AI to rebuild America’s defense industrial base and sustain its technological advantage.

VII. Additional References

Next-generation advanced manufacturing companies like Divergent have already developed and implemented advanced capabilities aligned with the AI-AM initiative. More information on these technologies is provided through the following references:

1. A *Gingrich 360* article, written by Newt Gingrich, highlighting how the integration of AI, 3D printing, materials science, and robotic assembly capabilities have enabled Divergent to build the “Factory of the Future” in Torrance, CA. (<https://gingrich360.com/2025/02/25/divergent-technologies-the-factory-of-the-future-is-here/>)
2. A video showing the process of Divergent’s AI-enabled technology being used to rapidly design, manufacture, and assemble an unmanned aerial vehicle structure. (<https://www.youtube.com/watch?v=iTrc7Z2Av4Q>)
3. A video of Divergent Founder Kevin Czingier describing how Divergent’s AI-enabled technology can be used to manufacture a nearly infinite range of products, like the “Star Trek Replicator.” (https://www.youtube.com/watch?v=Juo_YLcMKVY)